

DAMH ERT20 Free- t_0 Global-Archie Run Report

Generated from saved run artifacts

2026-06-15

Verdict

This run is a **failed sampler run with useful pre-crash exact OGS diagnostics**. The requested parametrisation was active: ERT t_0 was free, Archie parameters were fitted globally, and the open-niche boundary parameters were fixed. The run did not reach the planned stop at 2026-06-15 08:30 CEST. The launcher exited at 2026-06-14 22:59:45 CEST with exit code 139.

The immediate Python error was:

`ValueError: cannot reshape array of size 640 into shape (20,33)`

The failed evaluation appears to be `solver14_eval000003`. OGS itself finished with return code 0, but post-processing returned 640 ERT values, i.e. 20×32 , while the likelihood expected 660 values, i.e. 20×33 .

Configuration

Campaign	damh_ert20_t0_free_global_archie_su_until_20260615.0830_v1
Feature count	208
Observation count expected by likelihood	660
MPI size	16
Archie mode	global
Free ERT t_0	True
t_0 prior	mean 96.081672 d, std 10.00 d, bounds [66.081672, 106.081672] d
Open-niche boundary	fixed: shift 0.0 d, suction scale 1.0, suction offset 0.0 MPa
Dataset	dataset_arrays.npz from ert20_t0_free_global_archie_current
Warm checkpoint	ert20_theta_mlp_warm.pt from ert20_damh_t0_free_global_archie_seed

Run Outcome

Metric	Value
Result manifest written	2026-06-14T22:45:32+02:00
First exact candidate summary	2026-06-14T22:49:51+02:00
Last exact candidate summary	2026-06-14T22:59:38+02:00
Launcher exit code	139
Exact candidate summaries	44
Candidate status counts	{'evaluated': 44}
Likelihood history rows	4140
Likelihood calls	207
Likelihood history modes	{'global': 4140}
surrDAMH raw/sample/accepted rows flushed	1 / 1 / 1

Because only one surrDAMH sampling row was flushed, this run does **not** provide a meaningful posterior trace, accepted-sample ensemble, or convergence diagnostic. The reliable data products are the 44 exact OGS candidate summaries and their residual, Archie, and theta files.

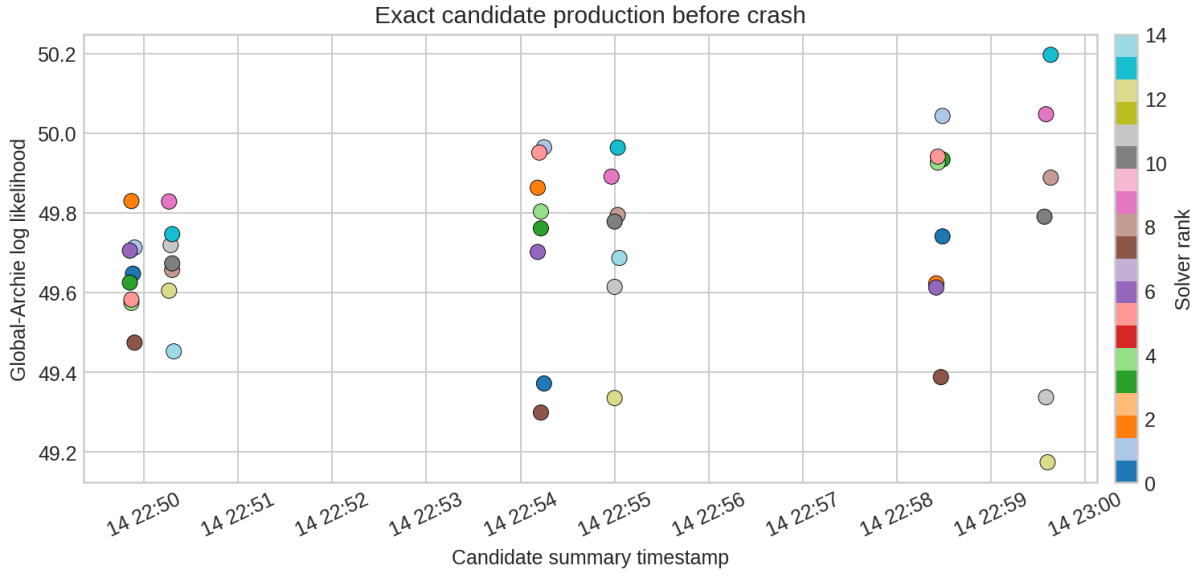
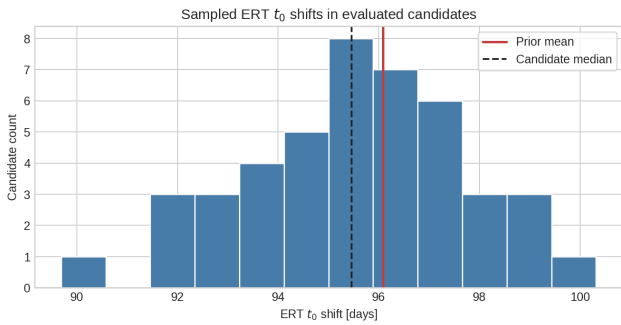


Figure 1: Exact-candidate production before the crash.

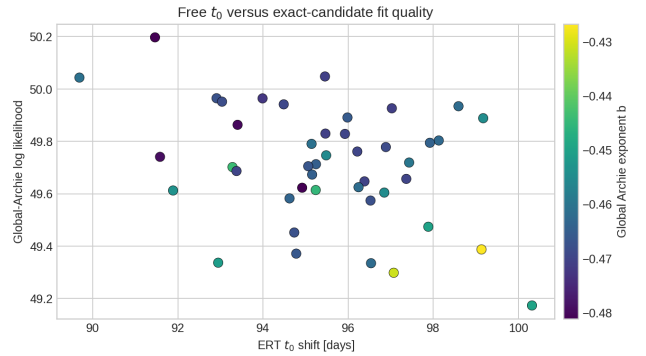
Exact Candidate Diagnostics

All summarized exact candidates used `archie_mode = global`, had `theta_output_count = 660`, and kept fixed open-niche boundary values.

Quantity	Min	Median	Mean	Max
Local Archie log likelihood	49.1736	49.7161	49.7104	50.1971
Local Archie MAE	0.08243	0.08368	0.08376	0.08546
Local Archie objective	0.33679	0.34882	0.34896	0.36238
Global Archie $\log_{10}(a)$	1.05834	1.07766	1.07875	1.11161
Global Archie exponent b	-0.48113	-0.46530	-0.46284	-0.42675
ERT t_0 shift [days]	89.6880	95.4619	95.4769	100.3196
$k_0 \log_{10}(k m^2)$	-21.2935	-20.9788	-20.9960	-20.5290
Homogeneous porosity	0.14018	0.14854	0.14798	0.15305
Bedding angle [deg]	138.7395	139.4510	139.4174	140.0752
q_{22}/q_{11}	0.43484	0.44953	0.45104	0.47322



(a) t_0 shift distribution.



(b) t_0 shift versus fit quality.

Figure 2: The free ERT t_0 shift was active in the valid candidates.

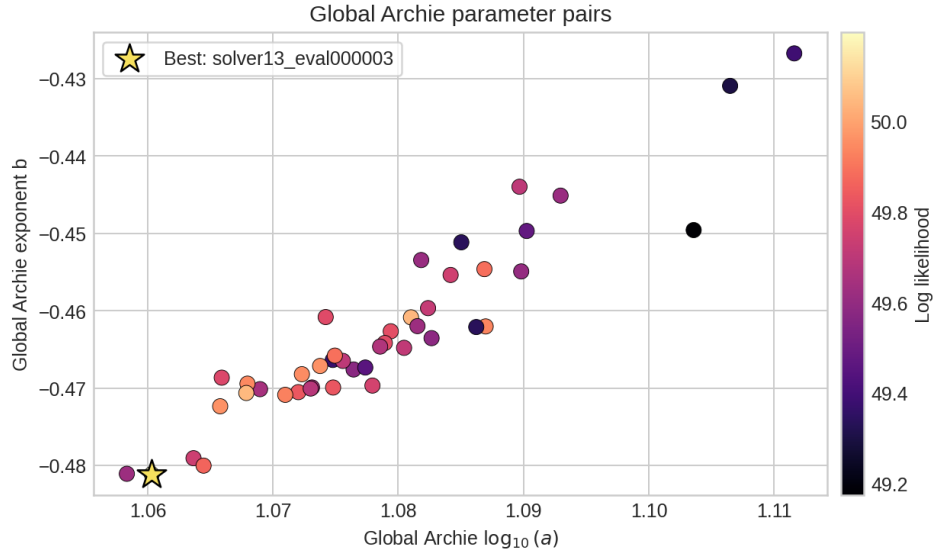


Figure 3: Global Archie parameter pairs fitted for each exact candidate.

Top Exact Candidates

Best exact candidate by global-Archie likelihood: `solver13_eval000003`.

Rank	Candidate	Log likelihood	MAE	t_0 shift [d]	Global $\log_{10}(a)$	Global b
1	<code>solver13_eval000003</code>	50.197105	0.082541	91.463850	1.060321	-0.481130
2	<code>solver09_eval000003</code>	50.047850	0.082492	95.456776	1.067875	-0.470627
3	<code>solver01_eval000003</code>	50.043519	0.083309	89.687973	1.081020	-0.460855
4	<code>solver01_eval000002</code>	49.964777	0.083288	92.909363	1.073765	-0.467137
5	<code>solver13_eval000002</code>	49.964145	0.082861	93.989998	1.065788	-0.472344

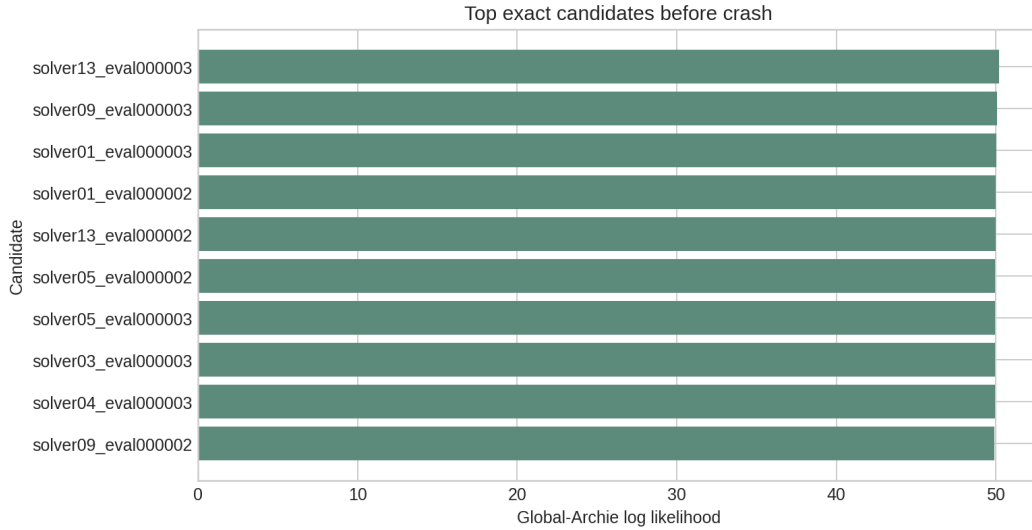


Figure 4: Top ten evaluated exact candidates before the crash.

Best Candidate Residual Diagnostics

`solver13_eval000003` has log likelihood 50.197105, MAE 0.082541, t_0 shift 91.463850 days, global $\log_{10}(a)$ 1.060321, and global b -0.481130.

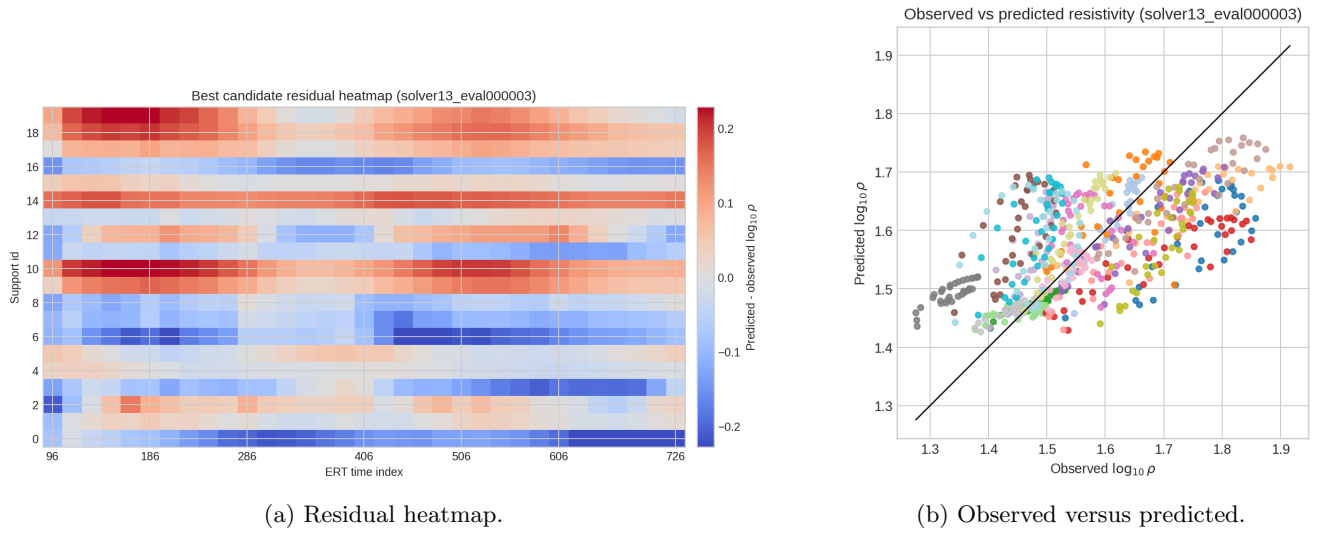


Figure 5: Best-candidate ERT residual checks.

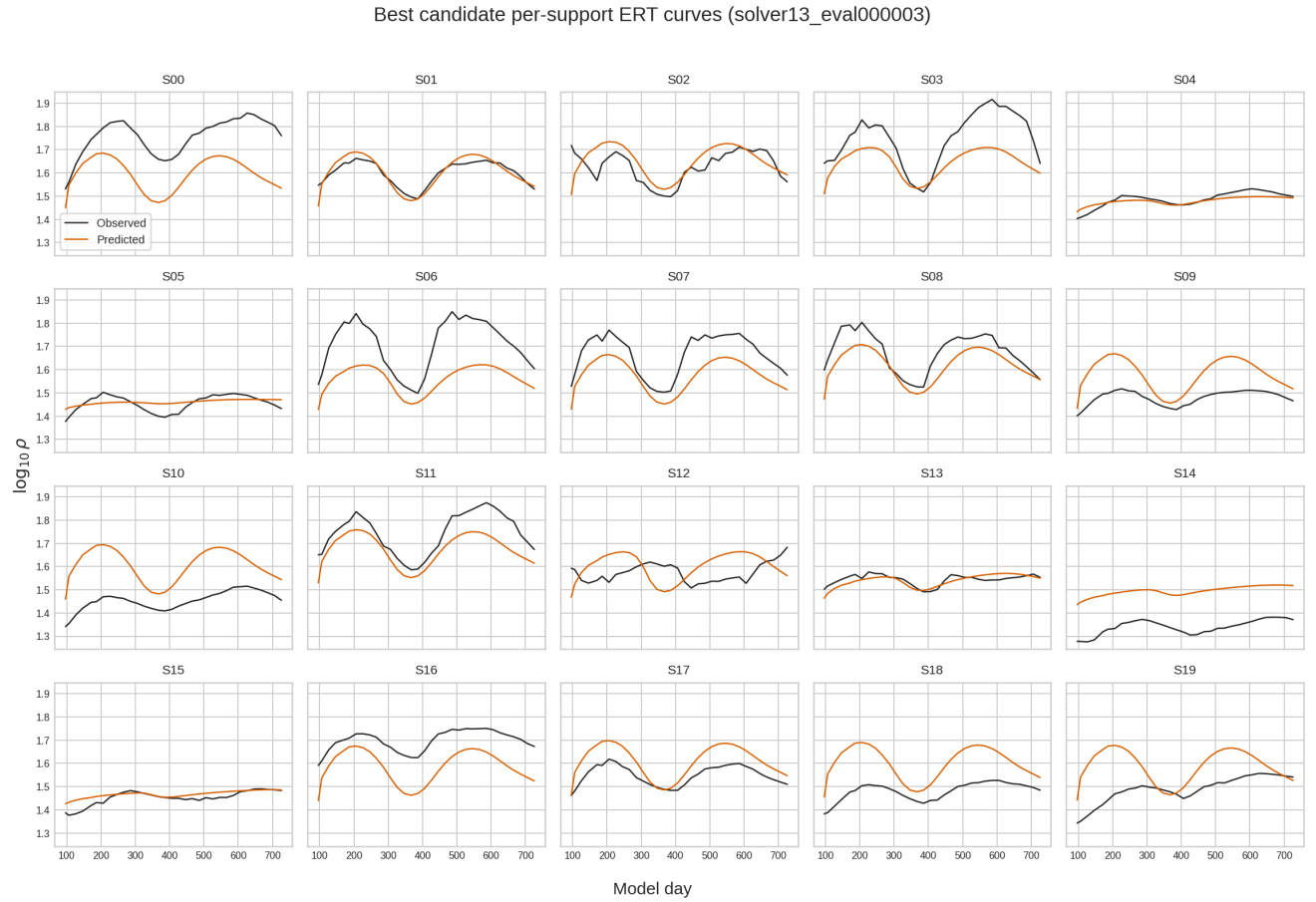


Figure 6: Observed and predicted ERT curves for all 20 supports for the best exact candidate.

Archie Spatial Comparison

The previous fixed-boundary deep run fitted Archie parameters independently per support, while this free- t_0 run fitted a single global Archie pair. The radial plot below uses support azimuth as the polar angle and separates the two ERT support depths into two traces. The dashed rings show the current best global Archie value.

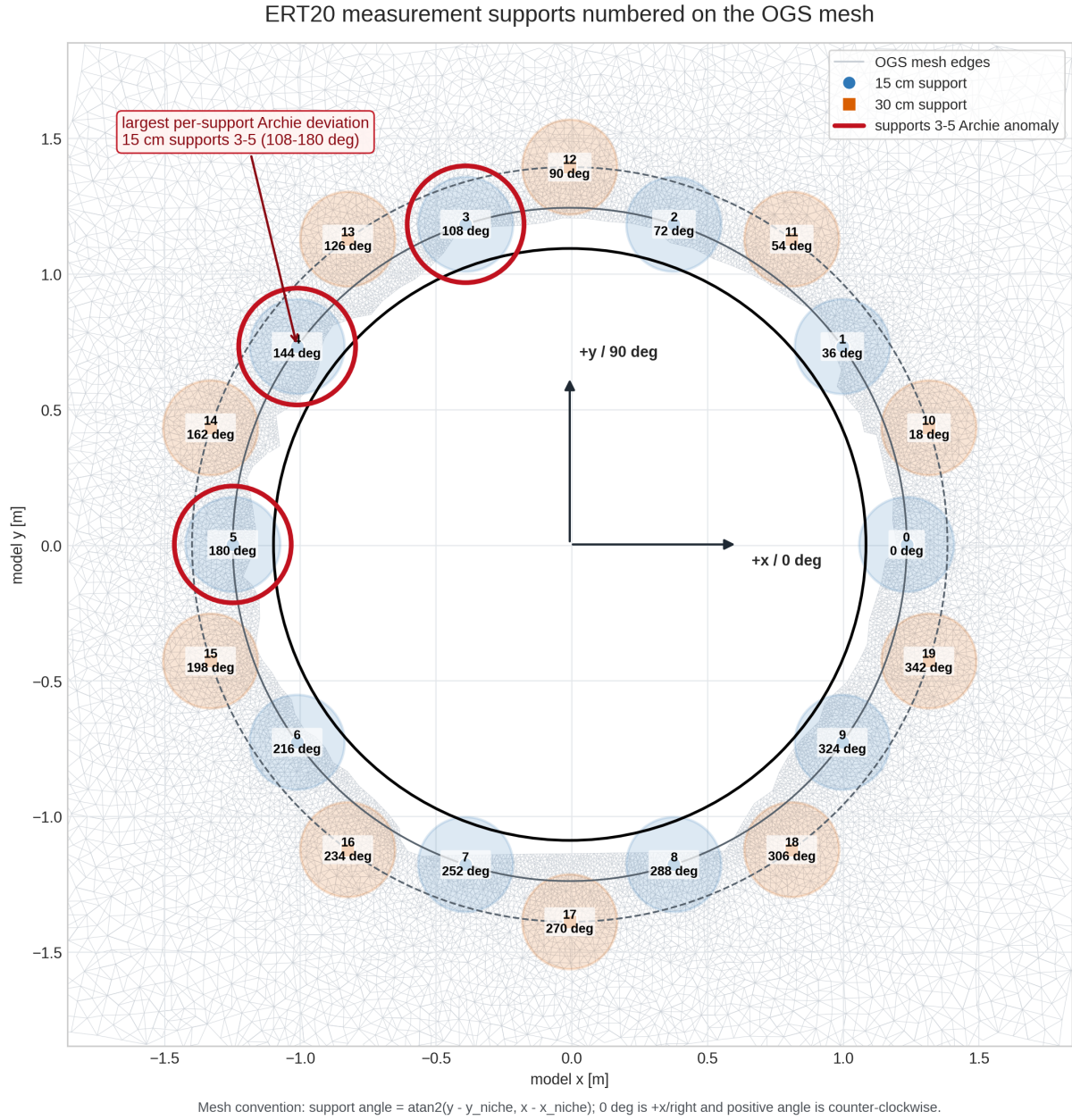


Figure 7: ERT20 measurement supports numbered on the OGS mesh. The red halos mark supports 3–5, the shallow 15 cm locations with the strongest previous per-support Archie departure.

Important spatial note for follow-up agents. The strongest departure from the global Archie pair is spatially localized and is dominated by the shallower 15 cm support trace, especially supports 3–5 at 108–180 degrees in the previous per-support fit. In the mesh-aligned coordinates this is the left/upper-left sector, not the bottom of the page; if a field-view niche drawing is rotated, translate the feature by support IDs/angles rather than by page position. A plausible hypothesis is that shallow/surface electrical current paths, contact effects, or near-boundary current leakage are breaking the simple OGS-theta-plus-Archie ERT model locally. This is not proven by the plot alone, but it is important enough to test explicitly before interpreting the global-Archie failure as a purely hydraulic or porosity-field problem.

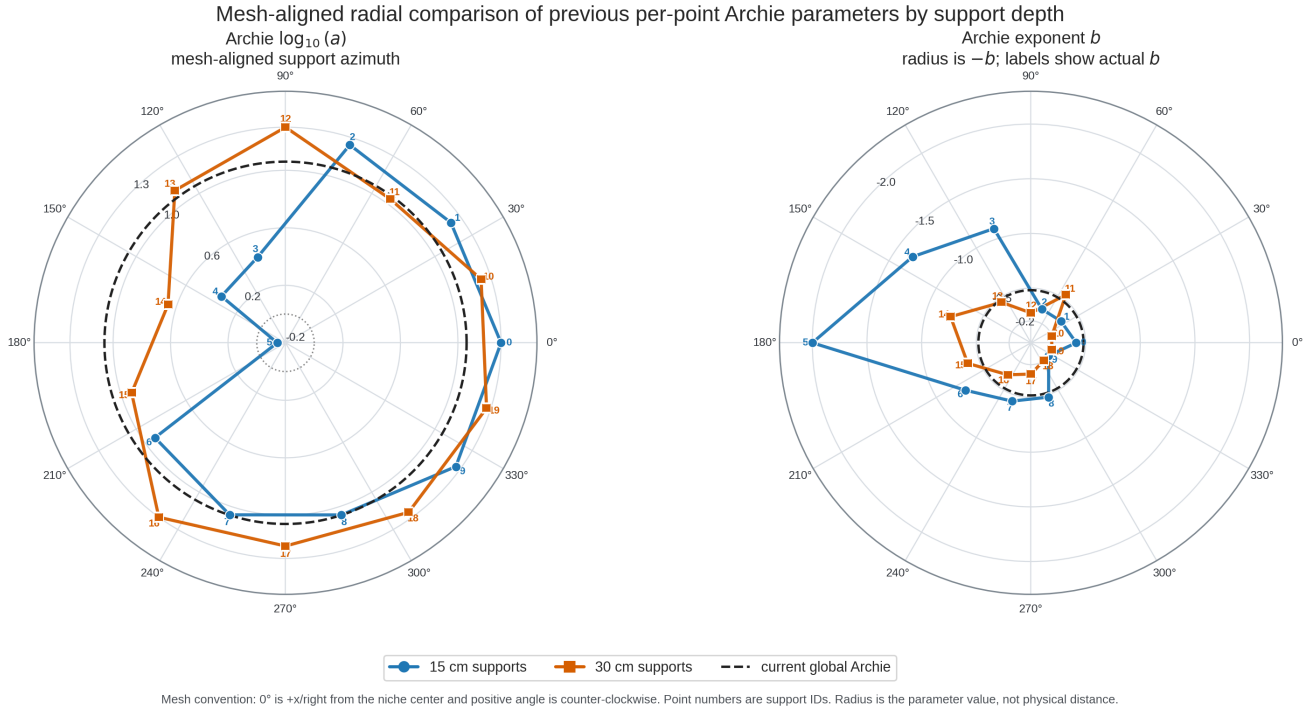


Figure 8: Mesh-aligned radial comparison of previous per-support Archie parameters against the current global Archie reference. The polar angle follows the support-table convention: 0° is positive mesh x from the niche center and angles increase counter-clockwise.

Failure Analysis

The core failure was a solver/likelihood contract mismatch:

$$\text{expected} = 20 \times 33 = 660, \quad \text{actual} = 20 \times 32 = 640.$$

Since the failed OGS run completed successfully, the likely failure surface is the ERT extraction and time-alignment step after OGS. A free t_0 perturbation can move one ERT measurement time outside the matched OGS output-time window, or outside the matching tolerance, leaving only 32 matched time points. The likelihood then attempted to reshape the vector into the fixed observed-data shape and crashed.

Scientific Use

Usable: the 44 evaluated exact candidate summaries; confirmation that free t_0 was sampled; confirmation that global Archie fitting was active; confirmation that the fixed open-niche boundary stayed fixed; rough early diagnostic distributions for fit quality and scalar parameters.

Not usable: posterior samples, acceptance-rate diagnostics, long-run convergence diagnostics, final surrogate-updated ensemble, or any morning-completion claim.

Recommended Relaunch Fix

Before relaunching, enforce that every candidate returns exactly $20 \times 33 = 660$ ERT values. If a shifted ERT date cannot be matched, either interpolate to the nearest valid OGS interval or reject the candidate with $-\infty$ likelihood before calling the Archie likelihood. Also add an explicit size guard in `get_observations` so one malformed candidate writes a failure summary instead of taking down MPI.